FLEET TEAM DIGEST

737MAX-FTD-32-18002

Issue Title : Main Landing Gear Wheel Fractures on Carbon Brake Wheels

Airplane Model	ATA	Minor Model(s)
737MAX	3245-11	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
737NG	9E991	

Originated Date	Last Revised Date	Created On
05/04/2018	09/18/2026	08/06/2026
Estimated Completion Date	**Next Update Date**	

Revision Description

Updated UTAS name to Collins Aerospace and provide the latest design changes to the wheel assemblies (Safran Landing Systems and Collins Aerospace) to mitigate wheel fractures.

Applicability

737-7/-8/-8200/-9 airplanes using wheel assemblies from either Safran Landing Systems or Collins Aerospace.

Description

There have been a small number of high energy fractures (ruptures) that have occurred on 737NG main gear wheels in service. The fractures have all occurred during pushback or taxi. The fractured segments of wheel were ejected inward toward the shock strut, causing damage to the brake and adjacent landing gear components, as shown in the attached photo.

These fractures have occurred on the following wheel part numbers:

Collins Aerospace P/N 3-1674
Safran Landing Systems P/N C20626200

While these events have occurred on 737NG airplanes, the same wheel part numbers are used on 737-7/-8/-8200/-9 models.

Background

The wheels consist of two halves held together by a ring of bolts, commonly known as tie bolts. Service experience has shown that if the preload in the tie bolts is less-than-optimal, the nuts may start backing out, or "unscrew" from the bolts as the wheel rolls under load. The mechanism that causes the nuts to unscrew is not well understood and has been difficult to reproduce on a dynamometer during lab testing. This nut loosening phenomenon appears unique to the wheel part numbers discussed in this article since this problem does not appear to occur on other wheels.

As the nuts unscrew from the bolts, the wheel halves start to separate under the influence of tire pressure and ground loads. Eventually the wheel halves flex to a sufficient amount that the wheel ruptures in the tubewell (see photo). Once the wheel ruptures, the fractured segment of wheel is ejected inward toward the main landing gear.

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The tendency for the nuts to loosen has been found to be associated with the following two issues:

- Bolt preload: Strict adherence to the bolt/nut tightening procedures has been shown to minimize the likelihood of bolt loosening.
- Bolt thread lubricant: Testing and in-service experience has shown that certain thread lubricants are more likely than others to result in nut loosening.

The referenced FTD articles discuss these two issues in more detail.

Status

Closed.

Interim Action

Boeing and the wheel suppliers have developed the following interim actions that have been shown by lab testing and/or in-service experience to be beneficial:

1) Both wheel suppliers have revised the CMM's for these wheel part numbers to require that the wheel tie bolt nuts be tightened with the torque-angle method (the torque only option has been removed).

2) Both wheel suppliers list approved tie bolt thread lubricants in their CMM's. Only thread lubricants shown in the CMM's should be used.

3) Both wheel suppliers have clarified the proper application of tie bolt thread lubricant in their CMM's (locations and lubricant volume). Both CMM's include photographs of proper lubricant application.

Final Action

Safran Landing Systems (SLS) has developed a new inner wheel half that incorporates a design feature at the main o-ring groove that allow the tire pressure to be gradually released should the tie bolts become loose in service.

Collins Aerospace has developed a new inner wheel half that is strengthened in the tubewell such that it prevents fracture if the tie bolts become loose.

Milestones

FTD closed.

Operator Action

Operators should ensure that their wheel MRO's have implemented the following actions to be in compliance with the wheel CMM's:

1) Ensure that only the specified tie bolt thread lubricants are used during wheel assembly.

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2) Ensure that the lubricant is applied per CMM guidance.
3) Ensure that the tie bolts are being tightened using the specified torque-angle method.

Operators using Safran Landing Systems (SLS) equipment:
SLS has created a new inner wheel half part number A39346. When using this new wheel half part number, the wheel assembly P/N C20746000 will become P/N C20746001. This wheel assembly was first delivered on a Boeing production airplane in August 2025 (LN 9259 and on). For the in-service fleet, SLS has created SB C20746-32-009 which contains information to install the new wheel half part number. This SLS Service Bulletin modifies the exiting wheel assembly P/N C20746000 to P/N C20746001. Please contact your local SLS Customer Support Manager (CSM) for more information. For wheel assembly P/N C20626200, Boeing has released Service Bulletin 737-32-1595 recommending replacement of wheel assembly P/N C20626200 and C20746000 with P/N C20746001.

Operators using Collins Aerospace equipment:
For wheel assembly P/N 3-1674, refer to Collins Aerospace Service Bulletin 3-1674-32-5. This Service Bulletin introduces a new wheel half assembly P/N 300-1039-2, which replaces inner wheel half assembly P/N 300-1039. This new inner wheel half introduces a "slow deflate" feature and also increases the fatigue life at the fuse plug port.

No operator action is required if using wheel assembly P/N 3-1709. This wheel contain an inner wheel half that is strengthened.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Component Maintenance Manual (CMM)	32-49-84 (Safran)		
Component Maintenance Manual (CMM)	32-42-32 (Collins)		
Component Maintenance Manual (CMM)	32-42-28 (Collins)		
Component Maintenance Manual (CMM)	32-49-83 (Safran)		
Fleet Team Digest Article (FTD)	737NG-FTD-32-12008		
Fleet Team Digest Article (FTD)	737NG-FTD-32-14004		
Service Bulletin (SB)	C20626-32-014 (Safran)		
Service Bulletin (SB)	C20626-32-016 (Safran)		
Service Bulletin (SB)	3-1674-32-5 (Collins)		
Service Bulletin (SB)	C20746-32-009 (Safran)		
Fleet Team Digest Article (FTD)	737MAX-FTD-32-21005		
Service Bulletin (SB)	737-32-1595		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Wheel Assy	S277A016-554	Safran Landing Systems	C20746001

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Wheel Assy	S277A016-353	Collins Aerospace	3-1709
Wheel Assy	S277A016-551	Safran Landing Systems	C20626200
Wheel Assy	S277A016-351	Collins Aerospace	3-1674
Wheel Assy	S277A016-553	Safran Landing Systems	C20746000

Related Categories

- Aging Airplane Issues : Structural issues or systems wear due to high flight cycles or high flight hours on the airplane
- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- New Equipment : Information on new parts available now or soon to be introduced
- Obsolescence : Information on parts that are obsolete, becoming obsolete, or have excessive lead times

Attachments:

a) Wheel Fracture

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